SOUTHEAST UNIVERSITY

Assignment on: configuration of multiple LANs with a DHCP server connected via a Router.

Course Code: ETE282.8

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Experiment Name: Configuration of multiple LAN'S with DHCP server connected via a Router.

Introduction:

In large-scale enterprise environments statically assigning IP addresses to hundreds of end-devices is inefficient and highly prone to administrative errors. The dynamic host configuration Protocol (DHCP) resolves this by automating the distribution of IP Parameters. In this lab experiment, a topology was designed utilizing cisco Packet tracer to implement multiple local area network (LANs) interconnected via a router. A dedicated centralized DHCP server was utilized to dynamically a allocate IP addresses to the end-devices across the network.

Objective:

- To design a multi-LAN topology utilizing switches a hub, and a central routing device.
- To assign and configure the 130.0.0.0 and 140.0.0.0 Class B network environment.

- To configure a dedicated server to act as a DHCP Pool for dynamic IP allocation.
- To successfully route traffic between two distinct separate subnets.
- To verify end-to-end inter-network connectivity using ICMP Packets.

Theoretical Background:

1. <u>Dynamic Host Configuration Protocol</u>: DHCP is a client/server Protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway. When a device configured as a DHCP client joins a network it broadcasts a DHCP discover message to locate a server, which then responds with a DHCP offer containing avilable IP Parameters.

2. <u>Intra-Network Routing</u>: To enable communication between two separate subnets 130.0.0.0 and 140.0.0.0 a layer 3 router is required. The router maintains active interfaces on both network and utilizes its internal routing table to

forward Packets across the separate broadcast domains.

3. Hubs vs Switches:

- Switch (Layer 2): forwards data Intelligently based on MAC addresses to specific destination Ports creating separate collision domains for each connected device.

- Hub (Layer 1): operates as a multi-Port repeater. it broadcasts incoming data out of all active Ports regardless of the destination resulting in a single shared collision domain.

4. Role of Centralized servers in a LAN: In a typical enterprise network servers Provide dedicated services to Client devices (PCs and Laptops):

- DHCP Server: Automates the assignments of IP addresses, subnet masks and default gateways ensuring no two devices have conflicting IP addresses.

- Application servers (HTTP/DNS/FTP): servers like server1 and server2 are typically assigned Static (unchanging) IP addresses so that client devices always know exactly where to reach them to access web Pages

4.1 Network Topology and Hardware Inventory:

The topology consists of two distinct network segments bridged by Router1:

- **Hardware Utilized:**
  - 1x cisco router → Router1.
  - 2x Cisco 2960-24TT switches
  - 1x hub
  - 3x PC's
  - 3x Laptops
  - 3x servers

- **Network Segmentation:**
  - Left LAN (130.0.0.0): connected to router Interface Fa0/0. It utilizes a cascaded switch archetecture (switch0 connected to switch1 to extend the network to multiple end-devices and the DHCP server.
  - Right LAN (140.0.0.0): connected to router interface Fa1/0. It utilizes a Physical Hub to distribute the connections to end-devices

4. Implementation and Configuration Steps:

1. Router Interface configuration: To facilitate routing between the two networks, the router requires static IP configurations on its FastEthernet interfaces to serve as the default gateways.

   - Access the router CLI and enter global configuration mode: configure terminal.
   - Configure fa0/0 (130.0.0.0 network gateway):
     - Interface fa0/0
     - IP address 130.0.0.1 255.255.0.0
     - no shutdown.
   - Configure fa1/0 (140.0.0.0 network gateway):
     - Interface fa1/0
     - IP address 140.0.0.1 255.255.0.0
     - no shutdown.

2. Server Configuration and Provisioning: Unlike the PCs and Laptops which receive dynamic IPs, all three servers in this topology requires manual, static IP configurations to ensure they remain reliably accessible.

1. <u>DHCP server configuration:</u>

- <u>Static IP assignment:</u> Assigned a static IP of $\boxed{130.0.0.2}$ Subnet: $\boxed{255.255.0.0}$ gateway: $\boxed{130.0.0.1}$

- <u>Service Activation:</u> Navigated to the services → DHCP tab and enabled the service.

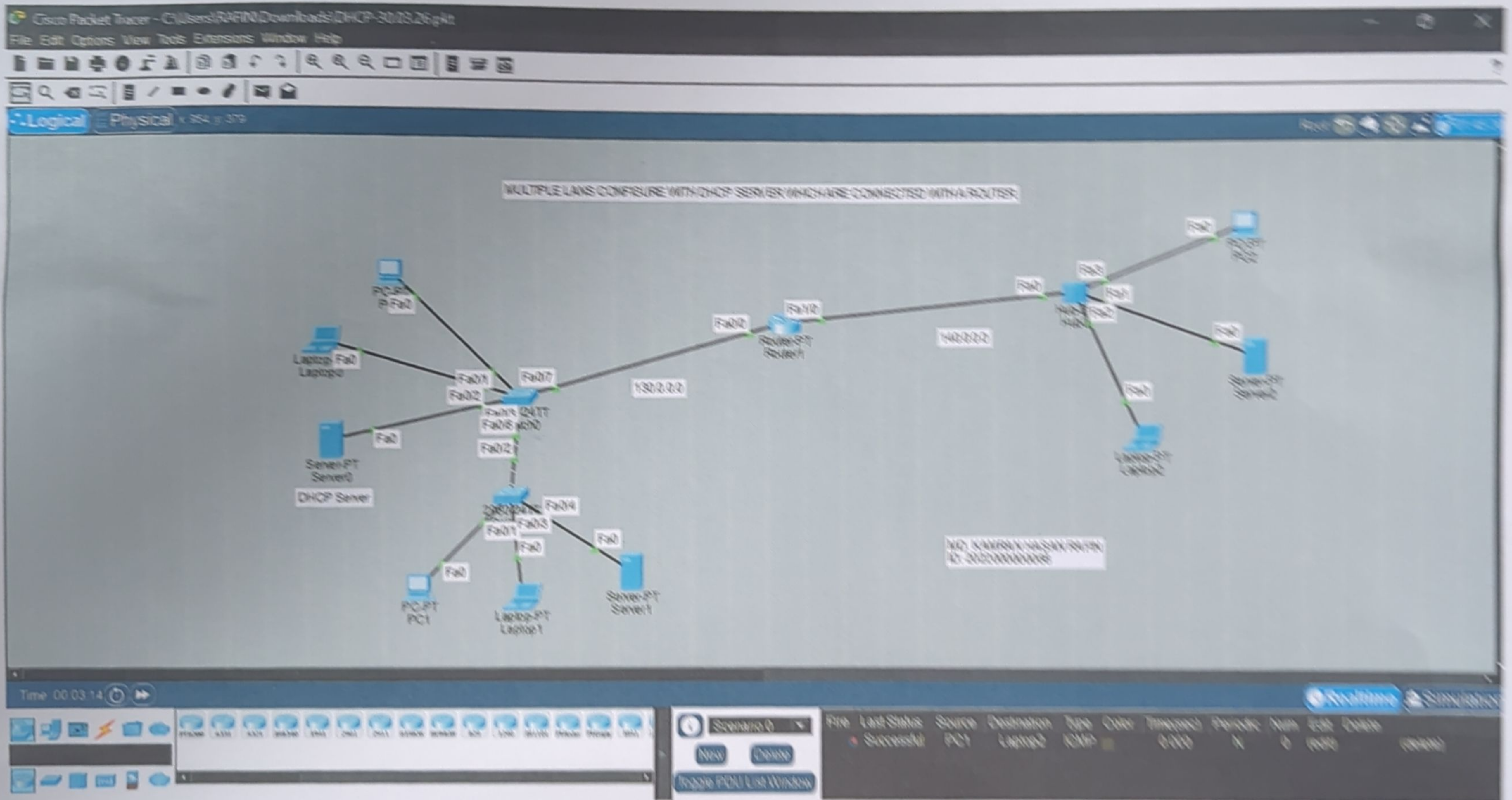
- <u>Pool configuration:</u> Programmed the server to hand out the $\boxed{130.0.0.1}$ default gateway and Configured the starting IP address for clients (e.g → starting at <u>130.0.0.10</u>) to Prevent it from handing out addresses already used by the router or other static servers.

2. <u>Application servers Configuration:</u>

- Server1 (on the 130.0.0.0 network) and server2 (on the 140.0.0.0 network) were assigned static IPs outside of the DHCP Client Pool.

- Their default gateways were strictly configured to match their respective router interfaces (130.0.0.1 and 140.0.0.1).

- HTTP services were verified as "ON" under the services tab, allowing any PC on either the left or right LAN to access centralized web Pages Proving that Application Layer


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
MULTIPLE LANS CONFIGURE WITH DHCP SERVER WHICH ARE CONNECTED WITH A ROUTER
Router-PT
Router1
Server-PT
Server0
DHCP Server
PC-PT
PC1
Laptop-PT
Laptop1
Server-PT
Server1
Fa0/1
Fa0/2
Fa0/7
Fa0
Time: 00:03:14
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC1 Laptop2 ICMP 0.000 N 0 (edit) (delete)

(Layer 7) services can successfully cross the Layer 3 router.

4. End-Device Configuration: Instead of manually assigning static IPs all connected PCs and Laptops were configured to obtain their configuration dynamically.

1. Open the device Properties and navigate to Desktop → IP Configuration.
2. Select the DHCP radio button instead of static.
3. The device successfully sent a DHCP request and populated its IP address, subnet masks and default gateway automatically from Server0.

4 Results And Conclusion:

The complex multi-LAN topology was successfully configured are deployed. The central DHCP server successfully leased IP addresses and gateway information to end devices across the network.

As evidenced by the Packet Tracer real-time simulation panel inter-network connectivity was successfull. An ICMP Ping test initiated from PC1 targeting Laptop2 returned a successfull status with a routing time of 0.000 seconds.

The lab exercise effectively demonstrated the scalability of modern networks. By implementing a DHCP server we successfully automated the tedious process of IP address management. Furthermore, the topology reinforced the functional differences between layer1 Hubs and layer2 switches while proving that a central layer 3 Router and dedicated servers can effectively manage and route traffic across dynamically addressed network segments.